

Approved	Issued
<i>S. Shiga</i>	<i>H. Okawa</i>

Consideration for crystal oscillation circuits

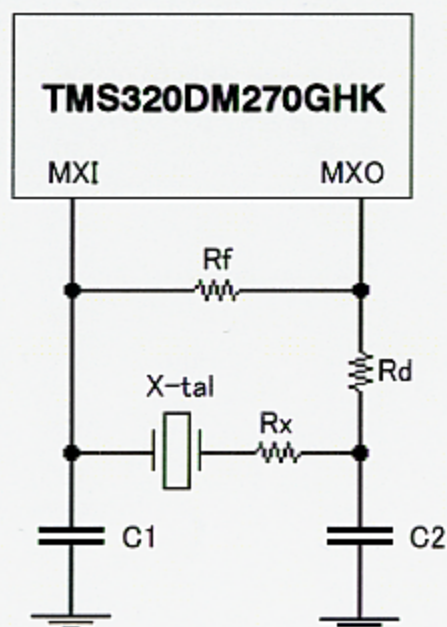
You will find typical data with regard to "the result of matching study of oscillation circuits and crystal units which is being performed at Kinseki". We hope this would be of any help in your future design work.

Remarks : The characteristics of the crystal oscillation circuits vary depending upon the circuit constants and actual mounting conditions , etc. Regarding the matching of crystal units and crystal oscillation circuits, you are requested to use with care upon confirmation. You are also requested to understand beforehand that the recommended constants and characteristics after following pages do not guarantee the characteristics of the product you design. Also, you are requested to refer to the "Remarks at the time of substrate design" as for the substrate design.

Result of study of crystal oscillation circuits

KINSEKI, LIMITED

Recommended circuit diagram:



	Model	CX-101F
	Size	3.2 × 2.5 × 0.60mm

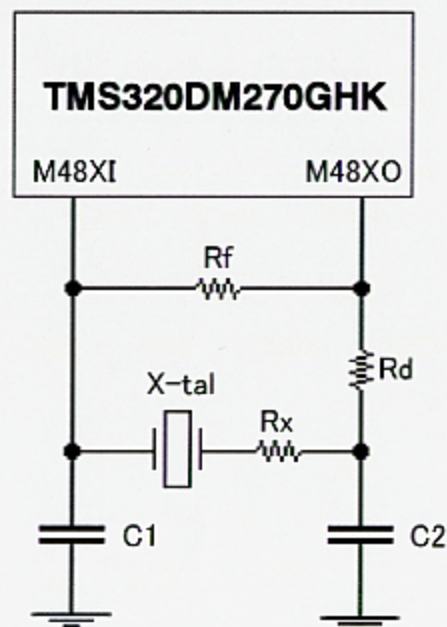
Result of study:

Model	CX-101F		
Frequency (kHz)	27000		
Overtone Order	Fundamental		
Working load capacitance (pF)	8.0		
<u>Recommended circuit constants</u>			
Rf	1MΩ		
Rd	0Ω		
Rx	150Ω		
C1	8pF		
C2	8pF		
<u>Characteristics of recommended constant</u>			
Negative resistance (Ω)	-691		
Circuit load capacitance (pF)	8.33		
Frequency stability (× 10 ⁻⁶)	-4.0		
Drive (μW)	45		
Kinseki internal control number	K1502-03A50-431		
Page of characteristics data	P4		

Result of study of crystal oscillation circuits

KINSEKI, LIMITED

Recommended circuit diagram:



	Model	CX-101F
	Size	3.2 × 2.5 × 0.60mm

Result of study:

Model	CX-101F		
Frequency (kHz)	48000		
Overtone Order	Fundamental		
Working load capacitance (pF)	8.0		
<u>Recommended circuit constants</u>			
Rf	1MΩ		
Rd	0Ω		
Rx	82Ω		
C1	3pF		
C2	3pF		
<u>Characteristics of recommended constant</u>			
Negative resistance (Ω)	-253		
Circuit load capacitance (pF)	6.94		
Frequency stability (× 10 ⁻⁶)	+26.1		
Drive (μW)	100		
Kinseki internal control number	K1502-03A51-431		
Page of characteristics data	P5		

Characteristics data in accordance with oscillation circuit constants
(Name of IC: TMS320DM270GHK)

KINSEKI, LIMITED

Evaluation data based on CX-101F/27000kHz CL:8.0pF

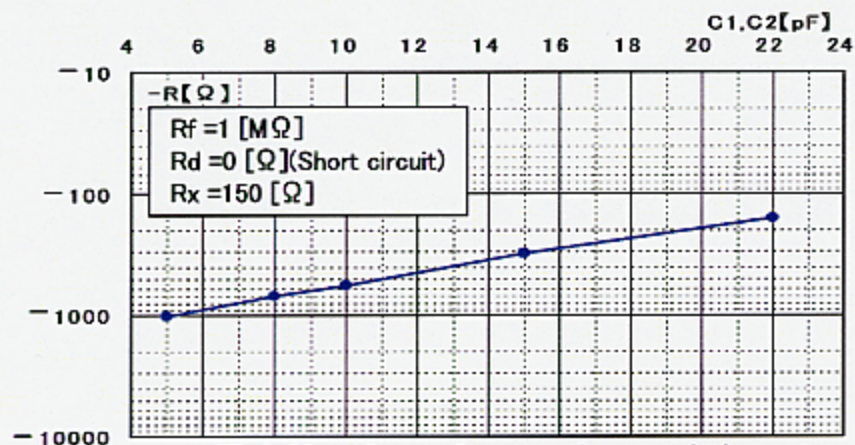


Fig1: Negative resistance characteristics of circuit to C1,C2

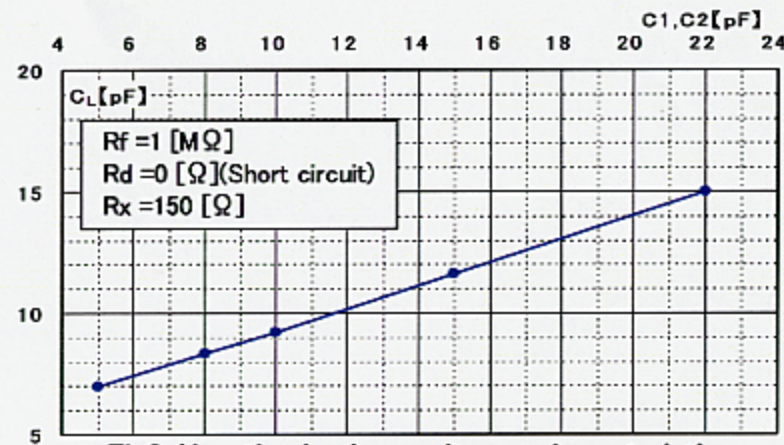


Fig2: Negative load capacitance characteristics of circuit to C1,C2

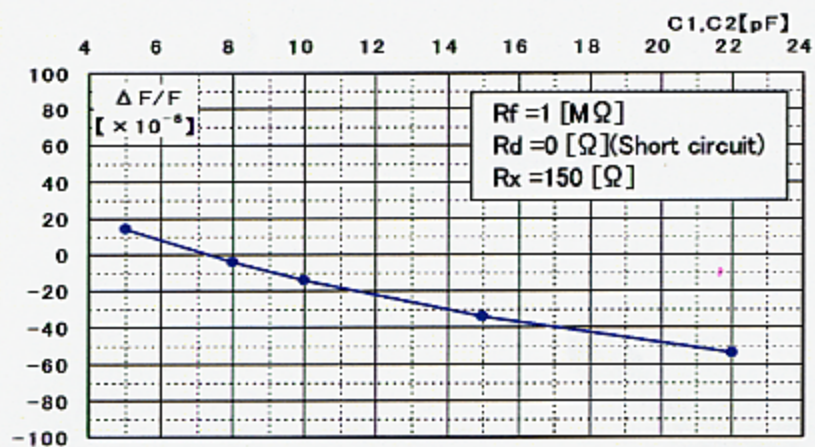


Fig3: Frequency stability characteristics to C1,C2

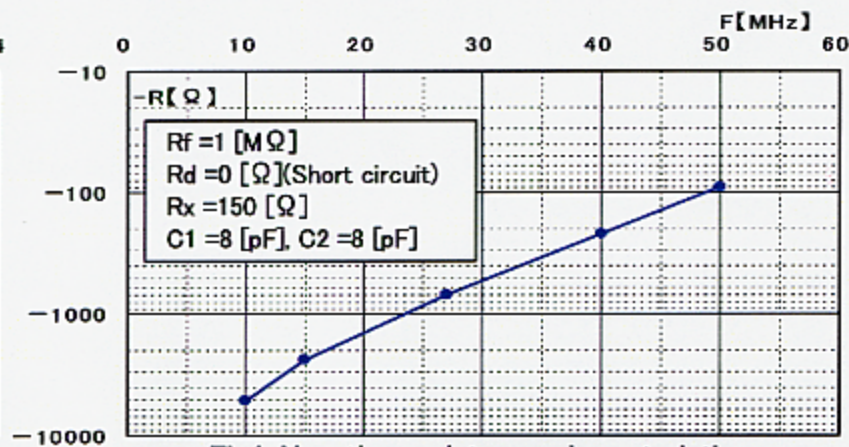


Fig4: Negative resistance characteristics of circuit to frequency

Characteristics data in accordance with oscillation circuit constants
(Name of IC: TMS320DM270GHK)

KINSEKI, LIMITED

Evaluation data based on CX-101F/48000kHz CL:8.0pF

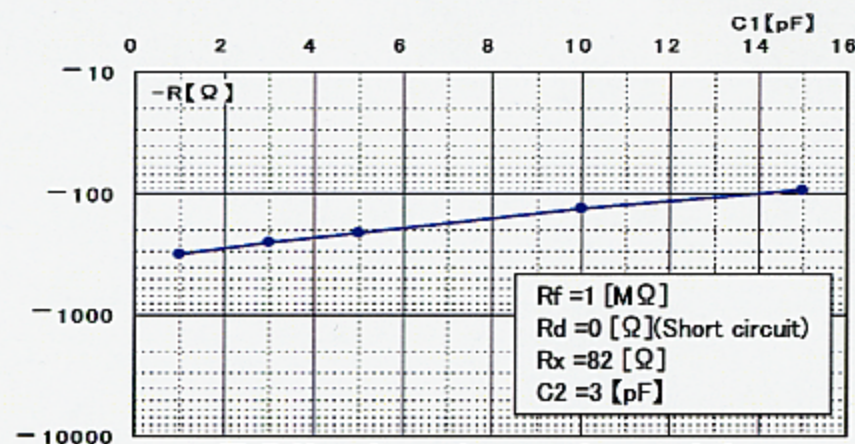


Fig5: Negative resistance characteristics of circuit to C1

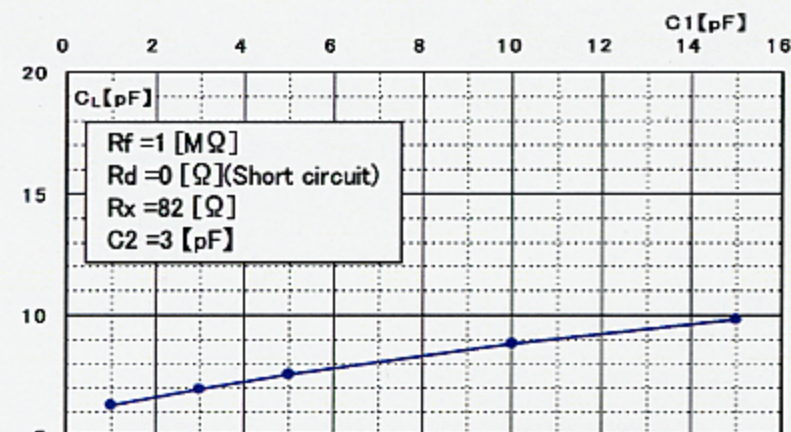


Fig6: Negative load capacitance characteristics of circuit to C1

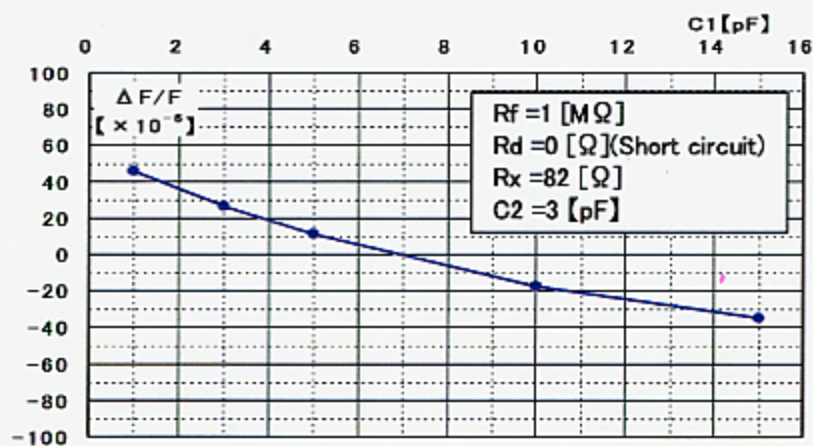


Fig7: Frequency stability characteristics to C1

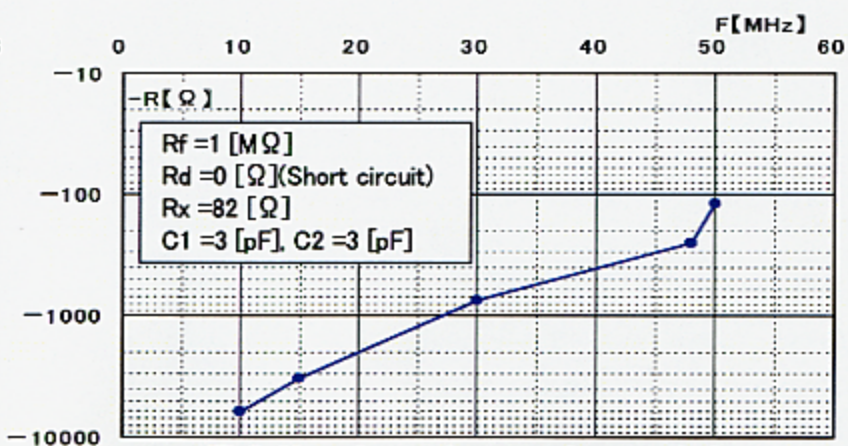


Fig8: Negative resistance characteristics of circuit to frequency

Remarks at the time of substrate design

KINSEKI, LIMITED

○ You are suggested to take note of followings when designing the oscillation circuit.

- Position the crystal oscillation circuit as much closer to the IC for oscillation as possible.
- The ground should not be arranged below the lower part of the crystal oscillation circuit.
- The wiring of the substrate should be as short as possible. (less than 2cm)
- Do not get close the patterns of the substrate each other connected to both sides of the crystal unit.
- Make the area as small as possible when the shield of the oscillation circuit is necessary.

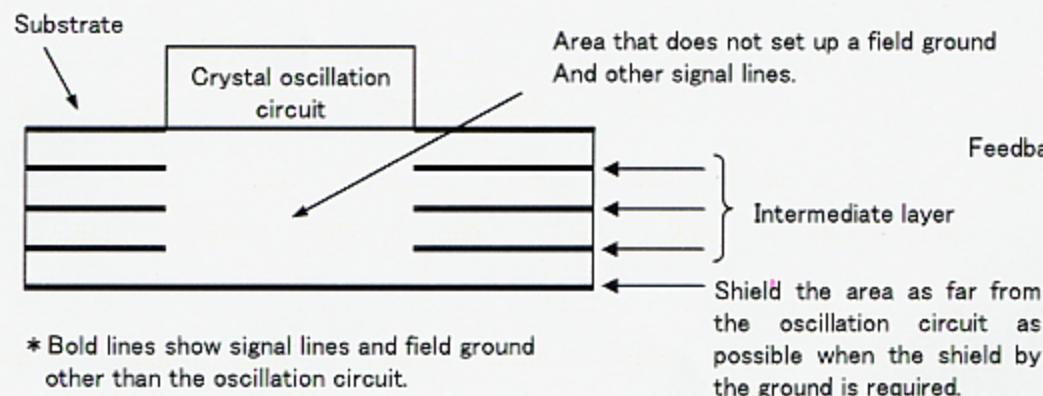


Fig.9:.. Cross section of substrate (in case of multilayer substrate)

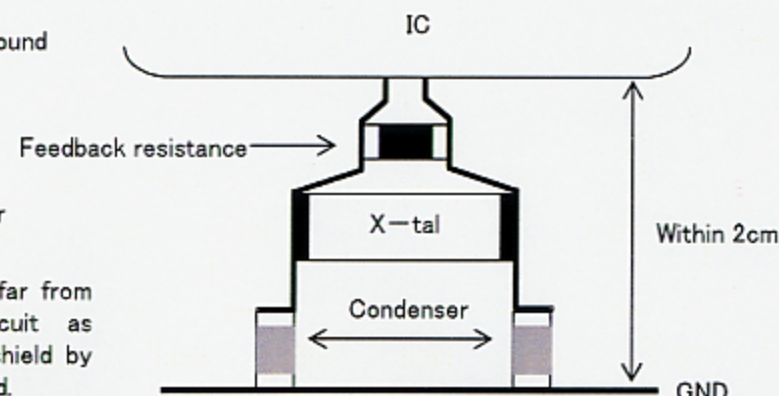


Fig.10: Plain figure of substrate

Inquiry

KINSEKI, LIMITED

As for any inquiry, you are suggested to contact as follows:

KINSEKI,LIMITED HEAD OFFICE

1-8-1,Izumihonchou,Komae-shi,Tokyo,201-8648 Japan
TEL (03) 5497-3117 FAX (03) 5497-3209

<http://www.kinseki.co.jp/>

E-mail: kanriks@kinseki.co.jp

AMERICAN KSS,INC. (KSS U.S.CORPORATE OFFICE)

2925 Bayview Drive Fremont,CA 94538,USA
TEL +1 (510) 661-4100 FAX +1 (510) 661-4101

<http://www.kss.com/>

E-mail: sales@kss.com

KINSEKI EUROPE GmbH

Wahler Park Wahlerstrasse 22D-40472 Dusseldorf Germany
TEL +49 (211) 368-150 FAX +49 (211) 368-1510

<http://www.kinseki.de/>

E-mail: sales@kinseki.de

KSS ELECTRONICS SINGAPORE PTE.,LTD

151 Lorong Chuan #03-07 New Tech Park(Lift Lobby G) Singapore 556741
TEL +65 6343-0400 FAX +65 6381-2300

<http://www.kss.com.sg/>

E-mail: info@kss.com.sg

KSS HONG KONG LTD.

UNIT 1801,18/F,TOWER III,Enterprise Square,No.9,Sheung Yuet Road,kowloon Bay, Kowloon1 Hong Kong
TEL (852) 2305-1080 FAX (852) 2305-1218

<http://www.kinseki.com.cn/>

E-mail: sales@kss.com.hk

KSS HONG KONG LTD. SHANGHAI REP. OFFICE

12F Majesty Building,138 Pudong Avenue,Shanghai, 1, CHINA 200120
TEL (86) 21-6859-9898 FAX (86) 21-5887-2542

<http://www.kinseki.com.cn/>

E-mail: sales@kinseki.com.cn

KSS ELECTRONICS (THAILAND) CO., LTD. BANGKOK SALES OFFICE

140/18 ITF TOWER II 11FL Simon Road, Bangrak Bangkok, 10500 THAILAND
TEL +66 (2) 634-3043 FAX +66 (2) 634-3042

E-mail: kssbkk@asiaaccess.net.th